JAMES' THEOREM AT EXTREME POINTS: LEFT-SYMMETRIC EXTREME POINTS CHARACTERIZE INNER PRODUCT SPACES

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Abstract. A classical theorem due to James and Day states that a real normed linear space of dimension at least three is an inner product space if and only if Birkhoff-James orthogonality is symmetric in it. We prove that it suffices to require symmetry at the extreme points of the unit ball. In finite dimensions: a finite-dimensional real Banach space of dimension at least three is an inner product space if and only if every extreme point of its unit ball is left-symmetric. In infinite dimensions, the same holds whenever extreme points are abundant in the unit ball, in the sense that every nonempty exposed face of the unit ball is the closed convex hull of its extreme points; in particular, for every real Banach space with the Radon-Nikodým property, or more generally with the Krein-Milman property, and for every reflexive space. Some such assumption is necessary, as c_0 shows. The key step is a geometric result valid in every real normed linear space: if a left-symmetric point is not smooth, then the unit sphere contains an open region which is ruled by segments parallel to that point. In particular, every left-symmetric point of a strictly convex space is smooth. For polyhedral spaces we obtain sharper information: a left-symmetric vertex is always parallel to at least four edges of the unit ball, and every polyhedral space of dimension at least three has at least four vertices which are not left-symmetric. The rhombic dodecahedron, which has exactly six such vertices, illustrates the results.

Status of this manuscript. This manuscript was written by an AI model (Claude, developed by Anthropic) under the guidance of Debmalya Sain. It is not meant for publication. At the time of writing, its results have not yet been verified by a human. Every proof has been checked by the author, but the reader should treat the results as unrefereed until a human verification is recorded here. The contributions of the witness and steerer are described in detail in Section 7.

Changes in Version 2 (29 October 2026). Version 2 adds the infinite-dimensional Theorem A′ and its proof (Section 4), answers Question 5.4 of Version 1 for reflexive spaces and for spaces with the Radon-Nikodým or the Krein-Milman property, replaces that question by Question 6.5, adds Example 6.3, and records in Section 7 the contribution of D. Sain which led to the new result. Sections 1–3 and 5–6 are otherwise unchanged in substance.

1. Introduction

The purpose of the present article is to show that the symmetry of Birkhoff-James orthogonality at the extreme points of the unit ball alone characterizes inner product spaces, in every finite dimension at least three and in a large class of infinite-dimensional Banach spaces, and to describe the geometric mechanisms by which symmetry fails at extreme points. Let us first establish the relevant notations and terminologies.

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Letters $\mathbb{X}$ denote real normed linear spaces. Let $B_{\mathbb{X}} = \{x \in \mathbb{X} : \|x\| \leq 1\}$ and $S_{\mathbb{X}} = \{x \in \mathbb{X} : \|x\| = 1\}$ be the unit ball and the unit sphere of $\mathbb{X}$, respectively, and let $\mathbb{X}^*$ denote the dual space of $\mathbb{X}$. For a nonempty convex set $C \subset \mathbb{X}$, $\operatorname{Ext} C$ denotes the collection of all extreme points of C. For a set $A \subset \mathbb{X}$, $\operatorname{co}(A)$, $\overline{\operatorname{co}}(A)$ and $\operatorname{aff}(A)$ denote the convex hull, the closed convex hull and the affine hull of A, respectively. For $x \in S_{\mathbb{X}}$, let

$$J(x) = \{f \in S_{\mathbb{X}^*} : f(x) = 1\}$$

be the collection of all supporting functionals at x. Each $f \in J(x)$ gives the supporting hyperplane $\{z \in \mathbb{X} : f(z) = 1\}$ of $B_{\mathbb{X}}$ at x, together with the parallel central hyperplane $\ker f$. A point $x \in S_{\mathbb{X}}$ is said to be smooth if $J(x)$ is a singleton, and $\mathbb{X}$ is said to be smooth if every point of $S_{\mathbb{X}}$ is smooth. $\mathbb{X}$ is said to be strictly convex if $\operatorname{Ext} B_{\mathbb{X}} = S_{\mathbb{X}}$, or equivalently, if $S_{\mathbb{X}}$ contains no nondegenerate segment.

For $f \in S_{\mathbb{X}^*}$, we write $F_f = \{z \in B_{\mathbb{X}} : f(z) = 1\}$. Note that $F_f \neq \emptyset$ if and only if f attains its norm. When F_f is nonempty, it is a closed face of $B_{\mathbb{X}}$, called the face exposed by f, and $\operatorname{Ext} F_f \subset \operatorname{Ext} B_{\mathbb{X}}$: indeed, if $x \in \operatorname{Ext} F_f$ and $x = \frac{1}{2}(a+b)$ with $a, b \in B_{\mathbb{X}}$, then $f(a) = f(b) = 1$, so $a, b \in F_f$ and $a = b$. For a face F, we denote by L_F the linear subspace parallel to $\operatorname{aff}(F)$, that is, $L_F = \operatorname{span}(F - F)$. In finite dimensions, F_f is nonempty for every $f \in S_{\mathbb{X}^*}$; moreover, every face F of $B_{\mathbb{X}}$ is compact and $F = \operatorname{co}(\operatorname{Ext} F)$.

Given $x, y \in \mathbb{X}$, x is said to be orthogonal to y in the sense of Birkhoff-James [1, 12], written as $x \perp_B y$, if

$$\|x + \lambda y\| \geq \|x\| \quad \text{for all } \lambda \in \mathbb{R}.$$

It is easy to observe that Birkhoff-James orthogonality is homogeneous, that is, $x \perp_B y$ implies $\alpha x \perp_B \beta y$ for all $\alpha, \beta \in \mathbb{R}$. Moreover, whether $x \perp_B y$ holds depends only on the restriction of the norm to $\operatorname{span}\{x, y\}$. We will repeatedly use the following characterization due to James.

Theorem 1.1 (James [12])**.** *Let $x \in S_{\mathbb{X}}$ and $y \in \mathbb{X}$. Then $x \perp_B y$ if and only if there exists $f \in J(x)$ such that $f(y) = 0$.*

Unlike the usual orthogonality in an inner product space, Birkhoff-James orthogonality is not symmetric in general. The following celebrated result of James and Day shows that symmetry is characteristic of inner product spaces.

Theorem 1.2 (James [13], Day [7])**.** *Let $\mathbb{X}$ be a real normed linear space with $\dim \mathbb{X} \geq 3$. Then Birkhoff-James orthogonality is symmetric in $\mathbb{X}$ if and only if $\mathbb{X}$ is an inner product space.*

To study the failure of symmetry locally, the following notions were introduced in [21].

Definition 1.3 ([21])**.** An element $x \in \mathbb{X}$ is said to be left-symmetric if $x \perp_B y$ implies $y \perp_B x$ for every $y \in \mathbb{X}$. An element $x \in \mathbb{X}$ is said to be right-symmetric if $y \perp_B x$ implies $x \perp_B y$ for every $y \in \mathbb{X}$.

Left-symmetric and right-symmetric points have since been studied extensively; we refer the readers to [23] for a complete characterization of such points in a Banach space, to [6] for the spaces ℓ_p^n, and to the monograph [15] for a detailed account.

The question studied in this article belongs to the program, initiated by Sain, Manna and Paul in [22], of studying global characterizations involving Birkhoff-James orthogonality in a local form. The classical Blanco-Koldobsky-Turnšek theorem [3, 14] states that a bounded linear operator which preserves Birkhoff-James orthogonality at every point is a scalar multiple of an isometry. In [22] the preservation of orthogonality by an operator at a single point was introduced and studied, and it was shown that this local study is connected with the extremal structure of the unit ball, leading to refinements of the Blanco-Koldobsky-Turnšek theorem on certain polyhedral spaces. The program was continued in [17] (directional preservation), in [16]

(approximate preservation), and in [18, 19], where it was shown that isometries can be detected by the preservation of Birkhoff-James orthogonality at the extreme points of the unit ball: in finite-dimensional spaces in [18], and more generally whenever the unit ball is the convex hull of its extreme points in [19, Theorem 2.27]. In the present article we carry this philosophy from the preservation of orthogonality by operators to the symmetry of orthogonality itself. Since every point of the unit sphere of a finite-dimensional space is a convex combination of extreme points, it is natural to ask whether the James-Day characterization survives when symmetry is required only at the extreme points of the unit ball.

Our first main result answers this question in the affirmative, and may be regarded as a version of Theorem 1.2 at extreme points.

Theorem A. *Let $\mathbb{X}$ be a real normed linear space with $3 \leq \dim \mathbb{X} < \infty$. Then $\mathbb{X}$ is an inner product space if and only if every extreme point of $B_{\mathbb{X}}$ is left-symmetric.*

In fact, we prove that if every extreme point of $B_{\mathbb{X}}$ is left-symmetric and $3 \leq \dim \mathbb{X} < \infty$, then $B_{\mathbb{X}}$ is strictly convex; Theorem A then follows from Theorem 1.2. The dimension bound is sharp: in the plane whose unit ball is a regular hexagon, every point of the unit sphere is left-symmetric. Left-symmetry cannot be replaced by right-symmetry either: the extreme points of the unit ball of ℓ_1^3 are right-symmetric (Example 6.2).

In infinite dimensions the unit ball may have few or no extreme points, and the condition of Theorem A then carries little or no information: the unit ball of c_0 has no extreme points at all (Example 6.3). D. Sain proposed (see Section 7) that Theorem A should remain true whenever extreme points are abundant, as they are in spaces with the Radon-Nikodým property. The following theorem confirms this. We say that $\mathbb{X}$ satisfies condition (E) if

$$\text{(E)} \qquad F_f = \overline{\text{co}}(\text{Ext}\, F_f) \quad \text{for every } f \in S_{\mathbb{X}^*} \text{ with } F_f \neq \emptyset.$$

A Banach space is said to have the Krein-Milman property (KMP) if every nonempty closed bounded convex subset is the closed convex hull of its extreme points. Every Banach space with the Radon-Nikodým property (RNP) has the KMP, by a theorem of Lindenstrauss (see [8, Chapter VII], [5]); in particular, every reflexive space and every separable dual space has the KMP. Every space with the KMP satisfies (E), and so does every reflexive space, directly from the Krein-Milman theorem, since F_f is then weakly compact.

Theorem A′. *Let $\mathbb{X}$ be a real Banach space with $\dim \mathbb{X} \geq 3$ satisfying condition* (E)*; for instance, let $\mathbb{X}$ have the Radon-Nikodým property, or the Krein-Milman property, or let $\mathbb{X}$ be reflexive. Then $\mathbb{X}$ is a Hilbert space if and only if every extreme point of $B_{\mathbb{X}}$ is left-symmetric.*

Theorem A is the special case $\dim \mathbb{X} < \infty$ of Theorem A′. We nevertheless give the finite-dimensional proof separately in Section 3, since it is completely elementary. The proof of Theorem A′ in Section 4 follows the same route; the only new ingredient is the Bishop-Phelps theorem [2], which replaces the norm attainment of functionals used in finite dimensions.

The key to Theorems A and A′ is the following geometric result, which holds in every real normed linear space and shows that corners of the unit sphere are an obstruction to left-symmetry.

Theorem D. *Let $\mathbb{X}$ be a real normed linear space and let $x \in S_{\mathbb{X}}$ be a left-symmetric point which is not smooth. Then $S_{\mathbb{X}}$ contains a nonempty relatively open subset which is a union of nondegenerate segments parallel to x. In particular, if $\mathbb{X}$ is strictly convex, then every left-symmetric point of $S_{\mathbb{X}}$ is smooth.*

We would like to mention that the smoothness of left-symmetric points has been observed earlier in specific operator spaces; for instance, it was shown in [4] that every nonzero left-symmetric operator in $\mathbb{B}(\ell_\infty^n, \ell_1^m)$ is smooth. Theorem D describes a general geometric mechanism behind such phenomena. Smoothness itself is no obstruction to left-symmetry: the unit vectors are smooth left-symmetric extreme points of ℓ_p^n, $1 < p < \infty$ (Remark 2.6). Theorem A shows, however, that in dimension at least three left-symmetry cannot hold at all extreme points at once, unless the space is an inner product space.

For polyhedral spaces, where every vertex is a non-smooth point, we obtain sharper and quantitative information.

Theorem C. *Let $\mathbb{X}$ be a real polyhedral Banach space with $\dim \mathbb{X} \geq 2$, and let x be a left-symmetric vertex of $B_\mathbb{X}$. Then $B_\mathbb{X}$ has at least two edges parallel to x. If $\dim \mathbb{X} \geq 3$, then $B_\mathbb{X}$ has at least four edges parallel to x.*

As an immediate consequence, the unit balls of ℓ_1^n and ℓ_∞^n, $n \geq 2$, have no left-symmetric vertices at all.

Theorem B. *Let $\mathbb{X}$ be a real polyhedral Banach space with $\dim \mathbb{X} \geq 3$. Then $B_\mathbb{X}$ has at least four vertices which are not left-symmetric.*

Left-symmetric vertices do exist in dimension three: the unit ball of the norm $\|y\| = \max_{i<j}(|y_i| + |y_j|)$ on $\mathbb{R}^3$ is the rhombic dodecahedron, and exactly six of its fourteen vertices are not left-symmetric (Example 6.1).

The proofs of Theorems A, B, C and D are elementary. They rest on a criterion for left-symmetric points (Lemma 2.1), which holds in every real normed linear space. The proof of Theorem B combines Theorem C with Euler's formula for three-dimensional polytopes, and then passes to higher dimensions by an inductive argument based on central sections. We would like to mention that, in a recent work [9], the classes of smooth and polyhedral spaces appear in connection with a different characterization of inner product spaces, namely the simultaneous zero inclusion property.

The article is organized as follows. In Section 2 we prove the criterion for left-symmetric points and Theorem D. Section 3 contains the proof of Theorem A, and Section 4 the proof of Theorem A$'$. Section 5 is devoted to polyhedral spaces and contains the proofs of Theorem B and Theorem C. In Section 6 we discuss examples and formulate some open questions. Section 7 describes in detail the contributions of the witness and steerer, and Section 8 describes how this manuscript was produced.

2. A criterion for left-symmetric points and non-smooth points

We begin with a criterion for left-symmetric points, which is valid in every real normed linear space.

Lemma 2.1. *Let $\mathbb{X}$ be a real normed linear space and let $x \in S_\mathbb{X}$ be left-symmetric. Let $y \in S_\mathbb{X}$, and suppose that there exists $g \in J(y)$ such that $g(x) > 0$. Then $f(y) \geq 0$ for every $f \in J(x)$.*

Proof. Suppose on the contrary that $f(y) < 0$ for some $f \in J(x)$. For $t \in [0,1]$, let $u(t) = (1-t)x + ty$. The function $t \mapsto f(u(t)) = (1-t) + tf(y)$ takes the value 1 at $t = 0$ and a negative value at $t = 1$. Therefore, $f(u(s)) = 0$ for some $s \in (0,1)$. Let us write $u = u(s)$ and $\beta = \|u\|$.

Since $B_\mathbb{X}$ is convex, $\beta \leq 1$. On the other hand, since $\|g\| = 1$, $g(y) = 1$, $g(x) > 0$ and $1 - s > 0$,

$$\beta \geq g(u) = (1-s)g(x) + s > s > 0.$$

In particular, $u \neq 0$, and we may consider $w = u/\beta \in S_{\mathbb{X}}$. Since $f \in J(x)$ and $f(w) = 0$, it follows from Theorem 1.1 that $x \perp_B w$. As x is left-symmetric, we have $w \perp_B x$. Once again by Theorem 1.1, there exists $h \in J(w)$ such that $h(x) = 0$. Then

$$\beta = \beta h(w) = h(u) = (1-s)h(x) + sh(y) = sh(y) \leq s,$$

since $\|h\| = \|y\| = 1$. This contradicts $\beta > s$, and completes the proof of the lemma. □

Remark 2.2. Geometrically, Lemma 2.1 states the following. If x lies strictly on the same side of the central hyperplane $\ker g$ as y, for some supporting hyperplane $\{g = 1\}$ of $B_{\mathbb{X}}$ at y, then y lies on the same closed side of $\ker f$ as x, for every supporting hyperplane $\{f = 1\}$ of $B_{\mathbb{X}}$ at x.

For $f_1, f_2 \in \mathbb{X}^*$ we write

$$W(f_1, f_2) = \{z \in S_{\mathbb{X}} : f_1(z) < 0 < f_2(z)\}.$$

We recall the following classical fact: for $z \in \mathbb{X} \setminus \{0\}$ and $v \in \mathbb{X}$, the one-sided directional derivatives of the norm are given by

$$\lim_{t \to 0+} \frac{\|z + tv\| - \|z\|}{t} = \max_{g \in J(z/\|z\|)} g(v), \qquad \lim_{t \to 0-} \frac{\|z + tv\| - \|z\|}{t} = \min_{g \in J(z/\|z\|)} g(v), \tag{1}$$

where the maximum and the minimum are attained by the weak* compactness of $J(z/\|z\|)$.

Theorem 2.3. *Let $\mathbb{X}$ be a real normed linear space and let $x \in S_{\mathbb{X}}$ be a left-symmetric point. Let $f_1, f_2 \in J(x)$ with $f_1 \neq f_2$. Then the following hold true:*

(a) $W(f_1, f_2)$ is a nonempty relatively open subset of $S_{\mathbb{X}}$, and $g(x) = 0$ for every $z \in W(f_1, f_2)$ and every $g \in J(z)$;

(b) for every $z \in W(f_1, f_2)$, the segment

$$\big[\, z - f_2(z)\, x,\ z - f_1(z)\, x \,\big]$$

is contained in $S_{\mathbb{X}}$. It is parallel to x, its length is $f_2(z) - f_1(z) > 0$, and it contains z in its relative interior.

Proof. (a) Since $f_1(x) = f_2(x) = 1$ and $f_1 \neq f_2$, the functionals f_1 and f_2 are linearly independent: indeed, if $f_2 = cf_1$, then evaluating at x gives $c = 1$. Hence the linear map $v \mapsto (f_1(v), f_2(v))$ from $\mathbb{X}$ to $\mathbb{R}^2$ is onto, and there exists $v \in \mathbb{X}$ with $f_1(v) < 0 < f_2(v)$. Then $v/\|v\| \in W(f_1, f_2)$. Clearly $W(f_1, f_2)$ is relatively open in $S_{\mathbb{X}}$.

Let $z \in W(f_1, f_2)$ and $g \in J(z)$. If $g(x) > 0$, then Lemma 2.1, applied to x, the point z and the functional $f_1 \in J(x)$, gives $f_1(z) \geq 0$, which is a contradiction. If $g(x) < 0$, then $-g \in J(-z)$ and $(-g)(x) > 0$, and Lemma 2.1, applied to x, the point $-z$ and the functional $f_2 \in J(x)$, gives $-f_2(z) = f_2(-z) \geq 0$, which is again a contradiction. Hence $g(x) = 0$.

(b) Let $C = \{v \in \mathbb{X} : f_1(v) < 0 < f_2(v)\}$. Then C is an open cone not containing 0, and $v \in C$ if and only if $v/\|v\| \in W(f_1, f_2)$. By (a) and (1), for every $v \in C$ both one-sided derivatives of the function $t \mapsto \|v + tx\|$ at $t = 0$ are equal to 0.

Now let $z \in W(f_1, f_2)$ and put $t_1 = -f_2(z)$, $t_2 = -f_1(z)$, so that $t_1 < 0 < t_2$. Since $f_1(x) = f_2(x) = 1$, we have $f_i(z + tx) = f_i(z) + t$ for $i = 1, 2$. Therefore $z + tx \in C$ for every $t \in (t_1, t_2)$. The function $\varphi(t) = \|z + tx\|$ is convex on $\mathbb{R}$, and by the previous paragraph, both of its one-sided derivatives vanish at every point of (t_1, t_2). A convex function whose right derivative vanishes at every point of an open interval is constant on that interval. Hence $\varphi(t) = \varphi(0) = 1$ for $t \in (t_1, t_2)$, and by continuity also for $t \in [t_1, t_2]$. This shows that the segment $\{z + tx : t_1 \leq t \leq t_2\}$ is contained in $S_{\mathbb{X}}$. It is parallel to x, its length is $(t_2 - t_1)\|x\| = f_2(z) - f_1(z) > 0$, and z corresponds to $t = 0 \in (t_1, t_2)$. □

Theorem 2.3 is Theorem D, since a non-smooth point x admits two distinct supporting functionals. It has the following consequences, which show that, under mild conditions, non-smooth points can never be left-symmetric.

Corollary 2.4. *Let $\mathbb{X}$ be a strictly convex real normed linear space. Then every left-symmetric point of $S_{\mathbb{X}}$ is smooth.*

Proof. The unit sphere of a strictly convex space contains no nondegenerate segment. The result now follows from Theorem 2.3. □

Corollary 2.5. *Let $\mathbb{X}$ be a real normed linear space, and let $x \in S_{\mathbb{X}}$ be such that $S_{\mathbb{X}}$ contains no nondegenerate segment parallel to x. If x is left-symmetric, then x is smooth.*

Remark 2.6. Smooth extreme points can be left-symmetric even when $\mathbb{X}$ is not an inner product space. For instance, let $\mathbb{X} = \ell_p^n$ with $1 < p < \infty$ and $n \geq 2$. The point e_1 is smooth and $J(e_1) = \{e_1^*\}$, so that $e_1 \perp_B y$ if and only if $y_1 = 0$. On the other hand, for $y \in S_{\ell_p^n}$, the unique supporting functional at y is $\sum_i |y_i|^{p-1} \operatorname{sgn}(y_i)\, e_i^*$, whose value at e_1 vanishes if and only if $y_1 = 0$. Hence e_1 is left-symmetric (see also [6]). Therefore, the conclusion of Corollary 2.4 cannot be improved to the failure of left-symmetry.

3. Proof of Theorem A

Throughout this section, $\mathbb{X}$ is a finite-dimensional real Banach space. We first study the faces of $B_{\mathbb{X}}$ of positive dimension under the assumption that their extreme points are left-symmetric.

Proposition 3.1. *Let $f \in S_{\mathbb{X}^*}$ be such that $F = F_f$ contains at least two points, and suppose that every extreme point of F is left-symmetric. Let $g \in S_{\mathbb{X}^*}$ take both a strictly positive and a strictly negative value on F, and write $M = \max_F g$ and $m = \min_F g$. Then the following hold true:*

(a) $F_g \subset \ker f$;
(b) $\|g + sf\| = 1$ for every $s \in [-M, -m]$;
(c) for every $y \in F_g$, the functionals $h_- = g - Mf$ and $h_+ = g - mf$ are two distinct elements of $J(y)$.

Proof. (a) Let $y \in F_g$, so that $g \in J(y)$. Suppose first that $f(y) < 0$. Let $x \in \operatorname{Ext} F$. Then x is an extreme point of $B_{\mathbb{X}}$, so that x is left-symmetric, and $f \in J(x)$. If $g(x) > 0$, then Lemma 2.1 gives $f(y) \geq 0$, a contradiction. Therefore, $g(x) \leq 0$ for every $x \in \operatorname{Ext} F$. Since $F = \operatorname{co}(\operatorname{Ext} F)$, we obtain $g \leq 0$ on F, which contradicts $M > 0$. If $f(y) > 0$, then we apply the same argument to the point $-y$ and the functional $-g \in J(-y)$, and obtain $g \geq 0$ on F, which contradicts $m < 0$. Hence $f(y) = 0$.

(b) Since $f = 1$ on F, for every $s \in \mathbb{R}$ the functional $g + sf$ takes the values $M + s$ and $m + s$ as its maximum and minimum on F. Hence, for every s in the open interval $I = (-M, -m)$, which contains 0, the functional $g + sf$ takes both a strictly positive and a strictly negative value on F. Let $s \in I$. Then $g + sf \neq 0$, and since $\mathbb{X}$ is finite-dimensional, there exists $y \in S_{\mathbb{X}}$ with $(g + sf)(y) = \|g + sf\|$. Applying (a) to the functional $(g + sf)/\|g + sf\|$, we obtain $f(y) = 0$, and therefore

$$\|g + sf\| = (g + sf)(y) = g(y) \leq 1.$$

Thus the convex function $\varphi(s) = \|g + sf\|$ satisfies $\varphi \leq 1$ on I and $\varphi(0) = 1$. Let $s \in I$, and choose $s' \in I$ such that 0 lies strictly between s and s', say $0 = \lambda s + (1 - \lambda)s'$ with $\lambda \in (0, 1)$. Then $1 = \varphi(0) \leq \lambda\varphi(s) + (1 - \lambda)\varphi(s') \leq 1$, which forces $\varphi(s) = 1$. Hence $\varphi = 1$ on I, and by continuity on $[-M, -m]$.

(c) Let $y \in F_g$. By (a), $f(y) = 0$, so that $(g + sf)(y) = g(y) = 1$ for every s. By (b), $\|h_-\| = \|h_+\| = 1$. Hence $h_-, h_+ \in J(y)$, and they are distinct since $f \neq 0$ and $M > m$. □

Lemma 3.2. *Let f, F, g, M, m, h_- and h_+ be as in Proposition 3.1, and suppose moreover that every extreme point of $B_{\mathbb{X}}$ is left-symmetric. Let y be an extreme point of F_g. Then:*

(a) for every $z \in F$ with $m < g(z) < M$, the face F contains a nondegenerate segment parallel to y which contains z in its relative interior;

(b) $y \in L_F$.

Proof. (a) The point y is an extreme point of the face F_g, hence an extreme point of $B_{\mathbb{X}}$, and therefore left-symmetric. By Proposition 3.1(c), h_- and h_+ are two distinct elements of $J(y)$. Let $z \in F$ with $m < g(z) < M$. Since $f(z) = 1$, we have $h_-(z) = g(z) - M < 0$ and $h_+(z) = g(z) - m > 0$, so that $z \in W(h_-, h_+)$. By Theorem 2.3(b), applied to the left-symmetric point y and the functionals $h_-, h_+ \in J(y)$, the segment $\sigma = [\, z - h_+(z)y,\ z - h_-(z)y\,]$ is contained in $S_{\mathbb{X}}$, is nondegenerate, is parallel to y, and contains z in its relative interior. By Proposition 3.1(a), $f(y) = 0$, so that $f = f(z) = 1$ on σ. Hence $\sigma \subset B_{\mathbb{X}} \cap \{f = 1\} = F$.

(b) Since F is convex and g takes the values $m < M$ on F, there exists $z \in F$ with $m < g(z) < M$. By (a), F contains a nondegenerate segment parallel to y, and therefore $y \in L_F$. □

Lemma 3.3. *Suppose that every extreme point of $B_{\mathbb{X}}$ is left-symmetric. Let $f \in S_{\mathbb{X}^*}$ be such that $F = F_f$ contains at least two points. Then* $\operatorname{span} F = \mathbb{X}$*, that is,* $\dim F = \dim \mathbb{X} - 1$.

Proof. We first note that there exists $g \in S_{\mathbb{X}^*}$ which takes both a strictly positive and a strictly negative value on F. Indeed, let $u, v \in F$ with $u \neq v$, choose $k \in \mathbb{X}^*$ with $k(u) > k(v)$, and let $g_0 = k - \frac{1}{2}(k(u) + k(v))f$. Then $g_0(u) > 0 > g_0(v)$, and $g = g_0/\|g_0\|$ has the required property.

Suppose on the contrary that $\operatorname{span} F \neq \mathbb{X}$. Then there exists a nonzero $h \in \mathbb{X}^*$ which vanishes on F. For $t \in \mathbb{R}$, the functional $g_t = g + th$ coincides with g on F, so that $g_t \neq 0$ and $\widehat{g}_t = g_t/\|g_t\|$ takes both a strictly positive and a strictly negative value on F. Let y_t be an extreme point of the nonempty compact face $F_{\widehat{g}_t}$. By Lemma 3.2(b), applied to $\widehat{g}_t$, we have $y_t \in L_F \subset \operatorname{span} F$, so that $h(y_t) = 0$. Therefore,

$$\|g_t\| = g_t(y_t) = g(y_t) \leq 1 \quad \text{for every } t \in \mathbb{R}.$$

On the other hand, choosing $w \in S_{\mathbb{X}}$ with $h(w) \neq 0$, we have $\|g_t\| \geq |g_t(w)| \geq |t|\,|h(w)| - 1$, which tends to infinity as $|t| \to \infty$. This contradiction shows that $\operatorname{span} F = \mathbb{X}$. Since $F \subset \{f = 1\}$, which does not contain 0, we have $\dim \operatorname{span} F = \dim F + 1$, and the proof is complete. □

We are now ready to prove our main structural result.

Theorem 3.4. *Let $\mathbb{X}$ be a real normed linear space with $3 \leq \dim \mathbb{X} < \infty$. If every extreme point of $B_{\mathbb{X}}$ is left-symmetric, then $\mathbb{X}$ is strictly convex.*

Proof. Suppose on the contrary that $S_{\mathbb{X}}$ contains a segment $[u, v]$ with $u \neq v$. Let $c = \frac{1}{2}(u + v)$ and $f \in J(c)$. Since $f(u) \leq 1$, $f(v) \leq 1$ and $\frac{1}{2}(f(u) + f(v)) = f(c) = 1$, we have $f(u) = f(v) = 1$, so that $F = F_f$ contains at least two points. By Lemma 3.3, $\dim F = \dim \mathbb{X} - 1 \geq 2$.

Let p be an extreme point of the compact convex set F, and let q be a relative interior point of F; then $q \neq p$, since $\dim F \geq 1$. Since $\dim \operatorname{aff}(F) \geq 2$, the affine functions on $\operatorname{aff}(F)$ vanishing at both p and q form a vector space of dimension $\dim \operatorname{aff}(F) - 1 \geq 1$, so there exists such a function a which is not identically zero, and hence not constant. As q is a relative interior point of F and $a(q) = 0$, a takes both a strictly positive and a strictly negative value on F. Since $\operatorname{aff}(F)$ is a hyperplane not containing 0, there is a unique $g_0 \in \mathbb{X}^*$ which coincides with a on $\operatorname{aff}(F)$. Let $g = g_0/\|g_0\|$, and let $M = \max_F g$, $m = \min_F g$. Then $m < 0 = g(p) < M$.

Let y be an extreme point of the face F_g. By Lemma 3.2(a), applied to the point $z = p$, the face F contains a nondegenerate segment which contains p in its relative interior. This contradicts the fact that p is an extreme point of F. Hence $\mathbb{X}$ is strictly convex. □

Theorem 3.5. *Let $\mathbb{X}$ be a real normed linear space with $3 \leq \dim \mathbb{X} < \infty$. Then $\mathbb{X}$ is an inner product space if and only if every extreme point of $B_{\mathbb{X}}$ is left-symmetric.*

Proof. The necessary part follows from Theorem 1.2. Conversely, suppose that every extreme point of $B_{\mathbb{X}}$ is left-symmetric. By Theorem 3.4, $\mathbb{X}$ is strictly convex, so that every point of $S_{\mathbb{X}}$ is an extreme point of $B_{\mathbb{X}}$, and hence left-symmetric. By homogeneity, Birkhoff-James orthogonality is symmetric in $\mathbb{X}$, and Theorem 1.2 shows that $\mathbb{X}$ is an inner product space. □

Remark 3.6. Theorem 3.4 and Theorem 3.5 fail in dimension two. A two-dimensional real normed space in which Birkhoff-James orthogonality is symmetric is called a Radon plane [20]. In a Radon plane every point of the unit sphere is left-symmetric, and there exist Radon planes which are not strictly convex: for example, the plane whose unit ball is a regular hexagon (see [23] and the references therein). In the proof of Theorem 3.4, the dimension assumption is used exactly once, to ensure that the face F has dimension at least two. We also note that there exist smooth Radon planes which are not Euclidean [20].

4. Proof of Theorem A′: abundance of extreme points in infinite dimensions

Throughout this section, $\mathbb{X}$ is a real Banach space, of arbitrary dimension. In the proof of Theorem A, finite-dimensionality was used in four ways:

(i) every face F of $B_{\mathbb{X}}$ satisfies $F = \mathrm{co}(\mathrm{Ext}\, F)$ (Proposition 3.1(a));
(ii) every $g \in S_{\mathbb{X}^*}$ attains its norm, so that $F_g \neq \emptyset$, and every nonempty face has an extreme point (Proposition 3.1(b), Lemma 3.3, Theorem 3.4);
(iii) the maximum and the minimum of a functional on a face are attained (Proposition 3.1);
(iv) faces have relative interior points, and $\mathrm{span}\, F$ is closed (Lemma 3.3, Theorem 3.4).

In the infinite-dimensional setting, (i) and the second half of (ii) are exactly what condition (E) provides. The first half of (ii) is replaced by the Bishop-Phelps theorem, (iii) by suprema and infima, and (iv) by the Hahn-Banach theorem. We use the Bishop-Phelps theorem in the following form.

Theorem 4.1 (Bishop-Phelps [2])**.** *Let $\mathbb{X}$ be a real Banach space. Then the set of norm attaining functionals is norm dense in $\mathbb{X}^*$.*

We also record an elementary perturbation fact. For a nonempty set $F \subset B_{\mathbb{X}}$ and $k \in \mathbb{X}^*$ we write $M_F(k) = \sup_F k$ and $m_F(k) = \inf_F k$.

Lemma 4.2. *Let $\emptyset \neq F \subset B_{\mathbb{X}}$, let $\ell, k \in \mathbb{X}^*$ and $\delta > 0$ with $\|k - \ell\| < \delta$. Then $|M_F(k) - M_F(\ell)| \leq \delta$ and $|m_F(k) - m_F(\ell)| \leq \delta$. In particular, if $M_F(\ell) > \delta$ and $m_F(\ell) < -\delta$, then k takes both a strictly positive and a strictly negative value on F, and $k \neq 0$.*

Proof. For $z \in F$ we have $|k(z) - \ell(z)| \leq \|k - \ell\|\, \|z\| < \delta$. The assertions follow by taking suprema and infima over F. □

Note that the threshold δ in Lemma 4.2 depends only on the numbers $M_F(\ell)$ and $m_F(\ell)$, and not on $\|\ell\|$. This will be important in the proof of Lemma 4.5.

Proposition 4.3. *Let $\mathbb{X}$ satisfy* (E)*. Let $f \in S_{\mathbb{X}^*}$ be such that $F = F_f$ contains at least two points, and suppose that every extreme point of F is left-symmetric. Let $g \in S_{\mathbb{X}^*}$ take both a strictly positive and a strictly negative value on F, and write $M = \sup_F g$ and $m = \inf_F g$. Then the following hold true:*

(a) $F_g \subset \ker f$;
(b) $\|g + sf\| = 1$ *for every* $s \in [-M, -m]$;
(c) *for every* $y \in F_g$, *the functionals* $h_- = g - Mf$ *and* $h_+ = g - mf$ *are two distinct elements of* $J(y)$.

Proof. Note that $-1 \le m < 0 < M \le 1$.

(a) Let $y \in F_g$, and suppose first that $f(y) < 0$. For every $x \in \operatorname{Ext} F$, x is an extreme point of $B_{\mathbb{X}}$, hence left-symmetric, and $f \in J(x)$; as in the proof of Proposition 3.1(a), Lemma 2.1 shows that $g(x) \le 0$. By (E), $F = \overline{\operatorname{co}}(\operatorname{Ext} F)$, and since g is continuous, $g \le 0$ on F. This contradicts $M > 0$. If $f(y) > 0$, the same argument applied to $-y$ and $-g \in J(-y)$ gives $g \ge 0$ on F, contradicting $m < 0$. Hence $f(y) = 0$.

(b) Let $I = (-M, -m)$, an open interval containing 0, and let $s \in I$. Put $\ell = g + sf$. Since $f = 1$ on F, we have $M_F(\ell) = M + s > 0$ and $m_F(\ell) = m + s < 0$. Let $0 < \delta < \min\{M + s, -(m + s)\}$. By Theorem 4.1, there is a norm attaining $k \in \mathbb{X}^*$ with $\|k - \ell\| < \delta$. By Lemma 4.2, $k \neq 0$ and $\widehat{k} = k/\|k\|$ takes both a strictly positive and a strictly negative value on F. Choose $y \in S_{\mathbb{X}}$ with $\widehat{k}(y) = 1$, that is, $y \in F_{\widehat{k}}$. Applying (a) to the functional $\widehat{k}$, we obtain $f(y) = 0$. Therefore

$$\|k\| = k(y) \le \ell(y) + \delta = g(y) + sf(y) + \delta = g(y) + \delta \le 1 + \delta,$$

and hence $\|\ell\| \le \|k\| + \delta \le 1 + 2\delta$. Letting $\delta \to 0$, we obtain $\|g + sf\| \le 1$ for every $s \in I$. The convexity argument in the proof of Proposition 3.1(b) now shows that $\varphi(s) = \|g + sf\|$ is identically 1 on I, and by continuity on $[-M, -m]$.

(c) Let $y \in F_g$. By (a), $f(y) = 0$, so that $h_\pm(y) = g(y) = 1$, and by (b), $\|h_-\| = \|h_+\| = 1$. Hence $h_-, h_+ \in J(y)$, and they are distinct since $f \neq 0$ and $M > m$. □

Lemma 4.4. *Let* $\mathbb{X}$ *satisfy* (E), *and suppose that every extreme point of* $B_{\mathbb{X}}$ *is left-symmetric. Let* f, F, g, M, m, h_- *and* h_+ *be as in Proposition 4.3, and suppose moreover that* g *attains its norm. Then* F_g *has an extreme point, and for every extreme point* y *of* F_g:

(a) *for every* $z \in F$ *with* $m < g(z) < M$, *the face* F *contains a nondegenerate segment parallel to* y *which contains* z *in its relative interior;*
(b) $y \in L_F$.

Proof. Since g attains its norm, $F_g \neq \emptyset$, and by (E), $F_g = \overline{\operatorname{co}}(\operatorname{Ext} F_g)$; in particular, $\operatorname{Ext} F_g \neq \emptyset$. Let $y \in \operatorname{Ext} F_g$. Then $y \in \operatorname{Ext} B_{\mathbb{X}}$, so y is left-symmetric. The proof of (a) is now word for word the proof of Lemma 3.2(a), with Proposition 3.1 replaced by Proposition 4.3; Theorem 2.3 holds in every normed space.

(b) Since $m < M$, there exist $u, v \in F$ with $g(u) < g(v)$. The midpoint $z = \frac{1}{2}(u + v) \in F$ satisfies $m \le g(u) < g(z) < g(v) \le M$. By (a), F contains a nondegenerate segment parallel to y, and therefore $y \in L_F$. □

Lemma 4.5. *Let* $\mathbb{X}$ *satisfy* (E), *and suppose that every extreme point of* $B_{\mathbb{X}}$ *is left-symmetric. Let* $f \in S_{\mathbb{X}^*}$ *be such that* $F = F_f$ *contains at least two points. Then* $\operatorname{span} F$ *is dense in* $\mathbb{X}$.

Proof. As in the proof of Lemma 3.3, there exists $g \in S_{\mathbb{X}^*}$ with $M_F(g) > 0 > m_F(g)$. Fix δ with $0 < \delta < \min\{M_F(g), -m_F(g)\}$.

Suppose on the contrary that $\operatorname{span} F$ is not dense in $\mathbb{X}$. By the Hahn-Banach theorem, there exists a nonzero $h \in \mathbb{X}^*$ which vanishes on F. For $t \in \mathbb{R}$, let $g_t = g + th$. Since $g_t = g$ on F, we have $M_F(g_t) = M_F(g)$ and $m_F(g_t) = m_F(g)$ for every t; in particular, the same δ serves for every t. By Theorem 4.1, there is a norm attaining $k_t \in \mathbb{X}^*$ with $\|k_t - g_t\| < \delta$. By Lemma 4.2, $k_t \neq 0$, and $\widehat{k}_t = k_t/\|k_t\|$ is a norm attaining element of $S_{\mathbb{X}^*}$ which takes both a strictly positive

and a strictly negative value on F. By Lemma 4.4, applied to $\widehat{k}_t$, the face $F_{\widehat{k}_t}$ has an extreme point y_t, and $y_t \in L_F \subset \operatorname{span} F$. Hence $h(y_t) = 0$, and

$$\|k_t\| = k_t(y_t) \leq g_t(y_t) + \delta = g(y_t) + \delta \leq 1 + \delta.$$

Consequently, $\|g_t\| \leq \|k_t\| + \delta \leq 1 + 2\delta$ for every $t \in \mathbb{R}$. On the other hand, choosing $w \in S_{\mathbb{X}}$ with $h(w) \neq 0$, we have $\|g_t\| \geq |t|\,|h(w)| - 1 \to \infty$ as $|t| \to \infty$. This contradiction completes the proof. □

Theorem 4.6. *Let $\mathbb{X}$ be a real Banach space with $\dim \mathbb{X} \geq 3$ satisfying* (E)*. If every extreme point of $B_{\mathbb{X}}$ is left-symmetric, then $\mathbb{X}$ is strictly convex.*

Proof. Suppose on the contrary that $S_{\mathbb{X}}$ contains a segment $[u, v]$ with $u \neq v$. By the Hahn-Banach theorem there is $f \in J(\frac{1}{2}(u+v))$, and as in the proof of Theorem 3.4, $u, v \in F = F_f$.

Choice of three points. By (E), F has an extreme point p. Since F has at least two points and $\mathbb{R}p \cap \{f = 1\} = \{p\}$, there is $a \in F \setminus \mathbb{R}p$. By Lemma 4.5, $\operatorname{span} F$ is dense in $\mathbb{X}$; since $\dim \mathbb{X} \geq 3$ and $\operatorname{span}\{p, a\}$ is a closed subspace of dimension two, $\operatorname{span} F \not\subset \operatorname{span}\{p, a\}$, and there is $b \in F \setminus \operatorname{span}\{p, a\}$. Thus p, a, b are linearly independent. This is the only place where the assumption $\dim \mathbb{X} \geq 3$ is used.

Choice of a functional. Define a linear functional g_0 on $\operatorname{span}\{p, a, b\}$ by $g_0(p) = 0$, $g_0(a) = 1$, $g_0(b) = -1$, and extend it to an element of $\mathbb{X}^*$ by the Hahn-Banach theorem. Let $g = g_0/\|g_0\|$. Then $g(p) = 0$, $g(a) > 0$ and $g(b) < 0$.

Passing to a norm attaining functional. Let $0 < \delta < \frac{1}{2}\min\{g(a), -g(b)\}$. By Theorem 4.1, there is a norm attaining $k \in \mathbb{X}^*$ with $\|k - g\| < \delta$. Since $\|a\| = \|b\| = \|p\| = 1$,

$$m_F(k) \leq k(b) < g(b) + \delta < -\delta < k(p) < \delta < g(a) - \delta < k(a) \leq M_F(k).$$

In particular $k \neq 0$, and the functional $\widehat{k} = k/\|k\| \in S_{\mathbb{X}^*}$ attains its norm, takes both a strictly positive and a strictly negative value on F, and satisfies $m_F(\widehat{k}) < \widehat{k}(p) < M_F(\widehat{k})$.

The contradiction. By Lemma 4.4(a), applied to $\widehat{k}$, an extreme point y of $F_{\widehat{k}}$ and the point $z = p$, the face F contains a nondegenerate segment which contains p in its relative interior. This contradicts $p \in \operatorname{Ext} F$. Hence $\mathbb{X}$ is strictly convex. □

Theorem 4.7. *Let $\mathbb{X}$ be a real Banach space with $\dim \mathbb{X} \geq 3$ satisfying* (E)*. Then $\mathbb{X}$ is a Hilbert space if and only if every extreme point of $B_{\mathbb{X}}$ is left-symmetric. In particular, this holds whenever $\mathbb{X}$ has the Radon-Nikodým property or the Krein-Milman property, and whenever $\mathbb{X}$ is reflexive.*

Proof. If $\mathbb{X}$ is a Hilbert space, Birkhoff-James orthogonality coincides with the usual orthogonality, and every point is left-symmetric. Conversely, by Theorem 4.6, $\mathbb{X}$ is strictly convex, so $\operatorname{Ext} B_{\mathbb{X}} = S_{\mathbb{X}}$, every point of $S_{\mathbb{X}}$ is left-symmetric, and by homogeneity Birkhoff-James orthogonality is symmetric in $\mathbb{X}$. Theorem 1.2, which holds in every real normed linear space of dimension at least three, shows that $\mathbb{X}$ is an inner product space; being complete, $\mathbb{X}$ is a Hilbert space. The final assertion follows from the discussion preceding Theorem A′ in the Introduction. □

Theorem 4.7 is Theorem A′.

Remark 4.8. (1) The proof uses (E) only for faces F_k exposed by norm attaining functionals k, and only through two consequences: $F_k = \overline{\operatorname{co}}(\operatorname{Ext} F_k)$ (in Proposition 4.3(a)), and $\operatorname{Ext} F_k \neq \emptyset$ (in Lemmas 4.4, 4.5 and Theorem 4.6). Condition (E) concerns only the exposed faces of the unit ball, and is therefore formally weaker than the Krein-Milman property.

(2) In a reflexive space every functional attains its norm, by James' theorem, so that Theorem 4.1 is not needed there: the proofs of Section 3 apply with max and min intact, with (E) obtained from the Krein-Milman theorem, and with Lemma 4.5 and the three-point construction of Theorem 4.6 replacing Lemma 3.3 and the relative interior argument.
(3) Some assumption of the type of (E) is necessary; see Example 6.3.

5. Polyhedral spaces

Throughout this section, $\mathbb{X}$ is a finite-dimensional real polyhedral Banach space, that is, $B_{\mathbb{X}}$ is a polytope. The extreme points of $B_{\mathbb{X}}$ are called vertices, and $\mathcal{V}$ denotes the set of vertices. A facet F of $B_{\mathbb{X}}$ is a maximal proper face, and the facet functional of F is the unique $f_F \in S_{\mathbb{X}^*}$ with $F = F_{f_F}$. We shall use the following standard facts on polytopes, all of which can be found in [11].

(P1) $B_{\mathbb{X}}$ is the intersection of the half-spaces $\{z : f_F(z) \le 1\}$, where F runs over all facets. Consequently, $\|z\| = \max_F f_F(z)$ for every $z \in \mathbb{X}$.
(P2) For $x \in \mathcal{V}$, $J(x) = \mathrm{co}\{f_F : F \text{ is a facet containing } x\}$.
(P3) Every point of $B_{\mathbb{X}}$ lies in the relative interior of exactly one face. A point in the relative interior of a proper face Φ lies in exactly those facets which contain Φ, and $\mathrm{aff}(\Phi)$ is the intersection of the hyperplanes $\{f_G = 1\}$ over the facets G containing Φ.
(P4) The graph formed by the vertices and the edges of $B_{\mathbb{X}}$ is connected. If $\dim \mathbb{X} \ge 3$, then every edge of $B_{\mathbb{X}}$ is contained in a two-dimensional face.
(P5) If $\dim \mathbb{X} = 3$, then every edge lies in exactly two facets, the graph formed by the facets, two facets being adjacent if they share an edge, is connected, and at a vertex lying on exactly three edges, these three edges and the three facets containing the vertex alternate around the vertex.

Since every vertex of a polytope is a non-smooth point, Theorem 2.3 applies to every left-symmetric vertex. The following elementary observation will be useful.

Lemma 5.1. *Let $x \in \mathcal{V}$. Then the following hold true:*

(a) if $z \in \mathbb{X}$ and $g(z) = 0$ for every $g \in J(x)$, then $z = 0$;
(b) the only point of $B_{\mathbb{X}}$ which lies in every facet containing x is x itself.

Proof. (a) Suppose on the contrary that $z \neq 0$. Let F be a facet containing x. Then $f_F \in J(x)$, and so $f_F(x \pm tz) = f_F(x) = 1$ for every $t \in \mathbb{R}$. Let E be a facet not containing x. Then $f_E(x) < 1$. Since there are only finitely many facets, there exists $t > 0$ such that $f_E(x \pm tz) < 1$ for every facet E not containing x. By (P1), both $x + tz$ and $x - tz$ belong to $B_{\mathbb{X}}$. Then x is the midpoint of a nondegenerate segment contained in $B_{\mathbb{X}}$, contradicting the fact that x is an extreme point.

(b) Let $z \in B_{\mathbb{X}}$ lie in every facet containing x. Then $f_F(z - x) = 0$ for every facet F containing x, and by (P2), $g(z - x) = 0$ for every $g \in J(x)$. By (a), $z = x$. □

Lemma 5.2. *Let $x \in \mathcal{V}$ be left-symmetric. Let*

$$\mathcal{V}^+(x) = \{y \in \mathcal{V} : f(y) \ge 0 \text{ for every } f \in J(x)\}, \qquad \mathcal{V}^-(x) = -\mathcal{V}^+(x).$$

Then the following hold true:

(a) $\mathcal{V}$ is the disjoint union of $\mathcal{V}^+(x)$ and $\mathcal{V}^-(x)$;
(b) if $y \in \mathcal{V}^+(x)$, then $g(x) \ge 0$ for every $g \in J(y)$; if $y \in \mathcal{V}^-(x)$, then $g(x) \le 0$ for every $g \in J(y)$.

Proof. (b) Let $y \in \mathcal{V}^+(x)$. By Lemma 5.1(a), applied at the vertex x with $z = y \neq 0$, there exists $f \in J(x)$ with $f(y) \neq 0$, so that $f(y) > 0$. Then $f(-y) < 0$. If possible, let $g(x) < 0$ for some $g \in J(y)$. Then $-g \in J(-y)$ and $(-g)(x) > 0$, and Lemma 2.1, applied to x, the point $-y$ and the functional $-g$, gives $f(-y) \geq 0$. This is a contradiction. The case $y \in \mathcal{V}^-(x)$ follows by applying the above argument to $-y \in \mathcal{V}^+(x)$, since $J(-y) = -J(y)$.

(a) Let $y \in \mathcal{V}$, and suppose that $f(y) < 0 < f'(y)$ for some $f, f' \in J(x)$. Lemma 2.1, applied to x, the point y and the functional f, shows that no $g \in J(y)$ satisfies $g(x) > 0$. Lemma 2.1, applied to x, the point $-y$ and the functional f', shows that no $g \in J(y)$ satisfies $g(x) < 0$. Hence $g(x) = 0$ for every $g \in J(y)$, and Lemma 5.1(a), applied at the vertex y, gives $x = 0$, which is absurd. Therefore, $\mathcal{V} = \mathcal{V}^+(x) \cup \mathcal{V}^-(x)$. If $y \in \mathcal{V}^+(x) \cap \mathcal{V}^-(x)$, then $f(y) = 0$ for every $f \in J(x)$, and Lemma 5.1(a) gives $y = 0$, which is again absurd. □

We note that in Lemma 5.2 only the vertex x is assumed to be left-symmetric; the vertex y is arbitrary.

Definition 5.3. Let $x \in \mathcal{V}$ be left-symmetric. An edge $[y, z]$ of $B_{\mathbb{X}}$ is said to be a crossing edge for x if $y \in \mathcal{V}^+(x)$ and $z \in \mathcal{V}^-(x)$. We denote by $\mathcal{C}(x)$ the set of all crossing edges for x. Since $\mathcal{V}^+(-x) = \mathcal{V}^-(x)$, we have $\mathcal{C}(-x) = \mathcal{C}(x)$.

Theorem 5.4. *Let* $\dim \mathbb{X} \geq 2$ *and let* $x \in \mathcal{V}$ *be left-symmetric. Then the following hold true:*

(i) every facet G containing a crossing edge for x satisfies $f_G(x) = 0$;
(ii) every crossing edge $[y, z]$ for x is parallel to x, that is, $z - y \in \mathbb{R}x$;
(iii) if $\dim \mathbb{X} \geq 3$*, then every two-dimensional face Φ of $B_{\mathbb{X}}$ containing a crossing edge for x contains at least two distinct crossing edges for x, which are parallel to each other;*
(iv) $\mathcal{C}(x)$ contains at least two edges, and if $\dim \mathbb{X} \geq 3$*, then $\mathcal{C}(x)$ contains at least four edges.*

In particular, $B_{\mathbb{X}}$ has at least two edges parallel to x, and at least four if $\dim \mathbb{X} \geq 3$.

Proof. (i) Let $[y, z] \subset G$ with $y \in \mathcal{V}^+(x)$ and $z \in \mathcal{V}^-(x)$. Then $f_G \in J(y) \cap J(z)$. By Lemma 5.2(b), $f_G(x) \geq 0$ and $f_G(x) \leq 0$.

(ii) By (P3), the line $\mathrm{aff}([y, z])$ through y and z is the intersection of the hyperplanes $\{f_G = 1\}$ over the facets G containing $[y, z]$. By (i), $f_G(x) = 0$ for every such G, so this line is invariant under translation by every multiple of x. Hence $z - y \in \mathbb{R}x$.

(iii) Let Φ be a two-dimensional face containing a crossing edge $[y, z]$. The polygon Φ has vertices both in $\mathcal{V}^+(x)$ and in $\mathcal{V}^-(x)$, and its boundary is a cycle of edges of $B_{\mathbb{X}}$ through all its vertices. Therefore, the boundary passes from one class to the other at least twice, and Φ contains at least two distinct crossing edges. By (ii), both of them are parallel to x.

(iv) The graph formed by the vertices and the edges of $B_{\mathbb{X}}$ is connected by (P4). Since $x \in \mathcal{V}^+(x)$ and $-x \in \mathcal{V}^-(x)$, Lemma 5.2(a) shows that some edge e_1 joins a vertex of $\mathcal{V}^+(x)$ to a vertex of $\mathcal{V}^-(x)$. Since $\mathcal{V}^-(x) = -\mathcal{V}^+(x)$, $-e_1$ is also a crossing edge, and $-e_1 \neq e_1$, as an edge does not contain 0. If $\dim \mathbb{X} \geq 3$, then by (P4) and (iii), e_1 lies in a two-dimensional face Φ, which contains a second crossing edge $e_2 \neq e_1$. The edges $-e_1$ and $-e_2$ lie in $-\Phi$. Let G be a facet containing Φ; then $f_G = 1$ on Φ and $f_G = -1$ on $-\Phi$, so $\Phi \cap (-\Phi) = \emptyset$. Hence $e_1, e_2, -e_1, -e_2$ are four distinct crossing edges for x. The final assertion now follows from (ii). □

Theorem 5.4 is Theorem C. It sharpens Theorem 2.3 in the polyhedral case, from segments in the unit sphere to edges of the unit ball, and yields the following simple test.

Corollary 5.5. *If no edge of $B_{\mathbb{X}}$ is parallel to a vertex of $B_{\mathbb{X}}$, then no vertex of $B_{\mathbb{X}}$ is left-symmetric. In particular, for $n \geq 2$, the unit balls of ℓ_1^n and of ℓ_∞^n have no left-symmetric vertices.*

Proof. The first assertion is immediate from Theorem 5.4. The vertices of $B_{\ell_\infty^n}$ are the points $(\pm 1, \dots, \pm 1)$, and its edges are parallel to the standard unit vectors $e_1, \dots, e_n$; for $n \geq 2$ no vertex is parallel to an edge. The vertices of $B_{\ell_1^n}$ are the points $\pm e_i$, and its edges are parallel to the vectors $e_i \pm e_j$ with $i \neq j$; again no vertex is parallel to an edge. □

We now turn to the number of vertices which are not left-symmetric. By Theorem 3.5, there is at least one antipodal pair of such vertices whenever $\dim \mathbb{X} \geq 3$. In dimension three, Euler's formula gives more precise information.

Proposition 5.6. *Let* $\dim \mathbb{X} = 3$. *Let* N_0 *be the number of vertices of* $B_{\mathbb{X}}$, *let* L *be the number of left-symmetric vertices, and let* Δ *be the number of triangular facets. Then*

$$N_0 - L \geq 2 + \tfrac{1}{2}\Delta.$$

Proof. Let N_1 and N_2 denote the numbers of edges and facets of $B_{\mathbb{X}}$, respectively. Since x is left-symmetric if and only if $-x$ is, the left-symmetric vertices form $L/2$ antipodal pairs. Let x, x' be left-symmetric vertices with $x' \notin \{x, -x\}$. By Theorem 5.4(ii), the edges in $\mathcal{C}(x)$ are parallel to x and the edges in $\mathcal{C}(x')$ are parallel to x'; since two unit vectors are parallel only if they are equal or opposite, $\mathcal{C}(x) \cap \mathcal{C}(x') = \emptyset$. For each antipodal pair $\{x, -x\}$ of left-symmetric vertices, the set $\mathcal{C}(x) = \mathcal{C}(-x)$ contains at least four edges, by Theorem 5.4(iv). Hence at least $2L$ edges are crossing edges for some left-symmetric vertex.

A triangle has no two parallel edges. Therefore, by Theorem 5.4(iii), applied with Φ a facet, no edge of a triangular facet is a crossing edge for a left-symmetric vertex. As every edge lies in exactly two facets, the triangular facets have at least $\frac{3}{2}\Delta$ edges altogether. Consequently,

$$N_1 \geq 2L + \tfrac{3}{2}\Delta. \tag{2}$$

On the other hand, counting the pairs (edge, facet containing the edge) gives

$$2N_1 \geq 3\Delta + 4(N_2 - \Delta) = 4N_2 - \Delta. \tag{3}$$

By Euler's formula, $N_0 - N_1 + N_2 = 2$, so that $N_2 = 2 - N_0 + N_1$. Substituting, we obtain $2N_1 \geq 8 - 4N_0 + 4N_1 - \Delta$, that is,

$$N_1 \leq 2N_0 - 4 + \tfrac{1}{2}\Delta. \tag{4}$$

Combining (2) and (4), we get $2L \leq 2N_0 - 4 - \Delta$, which is the desired inequality. □

Proposition 5.7. *Let* $\dim \mathbb{X} = 3$. *Then* $B_{\mathbb{X}}$ *has at least four vertices which are not left-symmetric.*

Proof. By Proposition 5.6, the vertices which are not left-symmetric form at least one antipodal pair. Suppose on the contrary that exactly one pair $\pm u$ of vertices is not left-symmetric, so that $L = N_0 - 2$. We use the notation of the proof of Proposition 5.6.

Step 1: equality holds throughout. The inequality of Proposition 5.6 gives $\Delta = 0$, and then (2) and (4) give $N_1 = 2L = 2N_0 - 4$. Therefore, every edge is a crossing edge for exactly one antipodal pair of left-symmetric vertices, and for each such pair $\{x, -x\}$ the set $\mathcal{C}(x)$ contains exactly four edges. Moreover, $N_2 = 2 - N_0 + N_1 = N_0 - 2$, so that $2N_1 = 4N_2$. As $\Delta = 0$, every facet has at least four edges, and (3) shows that every facet is a quadrilateral.

Step 2: every facet is a parallelogram. Let G be a facet. Each edge of G is a crossing edge for some left-symmetric vertex x, and by Theorem 5.4(iii), G then contains two distinct crossing edges for x, parallel to x. Two edges of a convex quadrilateral which are parallel must be opposite edges. Hence the four edges of G form two pairs of parallel opposite edges, and G is a parallelogram. We also note that, for a left-symmetric vertex x, the edges of $B_{\mathbb{X}}$ parallel to x

are exactly the four edges in $\mathcal{C}(x)$, since every edge is a crossing edge for exactly one antipodal pair and is parallel to it.

Step 3: $B_{\mathbb{X}}$ has exactly six facets. By Step 1, the sum of the degrees of the vertices is $2N_1 = 4N_0 - 8$. Since every vertex has degree at least three, at least eight vertices have degree three. As only $\pm u$ are not left-symmetric, there is a left-symmetric vertex x of degree three. Let e_1, e_2, e_3 be the edges at x, parallel to the vectors a, b, c, respectively. By (P5), the three facets at x are parallelograms $P_1 \supset e_1 \cup e_2$, $P_2 \supset e_2 \cup e_3$ and $P_3 \supset e_3 \cup e_1$.

The edge e_1 is a crossing edge for some left-symmetric vertex x_a, parallel to a, and by Step 2 the edges parallel to a are exactly the four edges in $\mathcal{C}(x_a)$. Each of them lies in exactly two facets, and each facet containing one of them is a parallelogram containing exactly two of them. Hence exactly four facets contain an edge parallel to a. The facets $P_1, P_3, -P_1, -P_3$ are four such facets, and they are distinct, because $x \in P_1 \cap P_3$ while $x \notin -P_1 \cup -P_3$. So these are all the facets containing an edge parallel to a. Similarly, the facets containing an edge parallel to b are exactly $\pm P_1, \pm P_2$, and those containing an edge parallel to c are exactly $\pm P_2, \pm P_3$.

Let $\mathcal{S} = \{\pm P_1, \pm P_2, \pm P_3\}$. Every edge of P_1 is parallel to a or to b, so the other facet containing such an edge belongs to $\mathcal{S}$; the same holds for every member of $\mathcal{S}$. Thus every facet sharing an edge with a member of $\mathcal{S}$ belongs to $\mathcal{S}$. By (P5), the facet graph is connected, and therefore $\mathcal{S}$ is the set of all facets.

Step 4: the contradiction. Let $f_i = f_{P_i}$, $i = 1, 2, 3$. By (P1), $B_{\mathbb{X}} = \{z : |f_i(z)| \leq 1,\ i = 1, 2, 3\}$. Since $B_{\mathbb{X}}$ is bounded, f_1, f_2, f_3 are linearly independent, and the vertices of $B_{\mathbb{X}}$ are exactly the points z with $|f_1(z)| = |f_2(z)| = |f_3(z)| = 1$. Every edge lies in two facets $\pm P_i$ and $\pm P_j$ with $i \neq j$, so it is parallel to a vector in $\ker f_i \cap \ker f_j$. The left-symmetric vertex x is parallel to the edges in $\mathcal{C}(x)$, and therefore $f_i(x) = 0$ for some i. This contradicts $|f_i(x)| = 1$, and completes the proof. □

In order to pass from dimension three to higher dimensions, we need the following notion and three lemmas.

Definition 5.8. Let F and E be facets of $B_{\mathbb{X}}$. We say that E is a belt facet of F if the facet functional f_F takes both a strictly positive and a strictly negative value on E. Equivalently, the central hyperplane $\ker f_F$, which is parallel to F, meets the relative interior of E.

We note that f_F cannot vanish identically on a facet E, because $\operatorname{aff}(E)$ does not contain 0 and therefore spans $\mathbb{X}$.

Lemma 5.9. *Let F be a facet of $B_{\mathbb{X}}$ which admits a belt facet E. Then some vertex of F is not left-symmetric.*

Proof. Suppose on the contrary that every vertex of F is left-symmetric. Fix a vertex x of F. Then $f_F \in J(x)$. If $f_E(x) > 0$, let us choose $y \in E$ with $f_F(y) < 0$. Then $f_E \in J(y)$, and Lemma 2.1 gives $f_F(y) \geq 0$, a contradiction. If $f_E(x) < 0$, let us choose $z \in E$ with $f_F(z) > 0$ and put $y = -z$. Then $y \in -E$, $f_F(y) < 0$, $-f_E \in J(y)$ and $(-f_E)(x) > 0$, so that Lemma 2.1 again leads to a contradiction. Therefore, $f_E(x) = 0$ for every vertex x of F. The vertices of F affinely span the hyperplane $\{f_F = 1\}$, which does not contain 0, and hence they linearly span $\mathbb{X}$. Thus $f_E = 0$, which contradicts $\|f_E\| = 1$. □

Lemma 5.10. *Let F be a facet of $B_{\mathbb{X}}$ which admits no belt facet, and let $\mathbb{W} = \ker f_F$, endowed with the restricted norm. Then every vertex of $B_{\mathbb{W}} = B_{\mathbb{X}} \cap \mathbb{W}$ is a vertex of $B_{\mathbb{X}}$. Consequently, every vertex of $B_{\mathbb{W}}$ which is not left-symmetric in $\mathbb{W}$ is a vertex of $B_{\mathbb{X}}$ which is not left-symmetric in $\mathbb{X}$.*

Proof. Let s be an extreme point of $B_{\mathbb{W}}$. If $\|s\| < 1$, then s is an interior point of $B_{\mathbb{X}}$, so that $s \pm tz \in B_{\mathbb{W}}$ for every $z \in \mathbb{W}$ and every sufficiently small $t > 0$, and s is not an extreme point

of $B_{\mathbb{W}}$. Hence $s \in S_{\mathbb{X}}$. Let Φ be the face of $B_{\mathbb{X}}$ containing s in its relative interior. Since Φ is a proper face, it is contained in some facet E of $B_{\mathbb{X}}$. As E is not a belt facet of F, either $f_F \geq 0$ on E or $f_F \leq 0$ on E; without loss of generality, let us assume that $f_F \geq 0$ on Φ. If $f_F(p) > 0$ for some $p \in \Phi$, then, since s is a relative interior point of Φ, the point $q = s + t(s-p)$ belongs to Φ for all sufficiently small $t > 0$, and $f_F(q) = -tf_F(p) < 0$, which is impossible. Therefore $f_F = 0$ on Φ, that is, $\Phi \subset B_{\mathbb{W}}$. Since s is an extreme point of $B_{\mathbb{W}}$ lying in the relative interior of Φ, we obtain $\Phi = \{s\}$, and s is a vertex of $B_{\mathbb{X}}$.

For the second assertion, let s be a vertex of $B_{\mathbb{W}}$ which is left-symmetric in $\mathbb{X}$. Since Birkhoff-James orthogonality between two elements depends only on the norm restricted to their linear span, s is left-symmetric in $\mathbb{W}$. This proves the lemma. □

Lemma 5.11. *Let $u \in \mathcal{V}$, and suppose that every facet of $B_{\mathbb{X}}$ contains u or $-u$. Then no vertex of $B_{\mathbb{X}}$ other than $\pm u$ is left-symmetric.*

Proof. Let $v \in \mathcal{V} \setminus \{u, -u\}$. By Lemma 5.1(b), not every facet containing v contains u, since otherwise u would lie in every facet containing v and hence $u = v$. So there is a facet G_1 containing v but not u; by hypothesis, $-u \in G_1$, that is, $f_{G_1}(u) = -1$. In the same way, since $-u \neq v$, there is a facet G_2 containing v but not $-u$, so that $u \in G_2$ and $f_{G_2}(u) = 1$. Now $f_{G_1}, f_{G_2} \in J(v)$ take values of opposite signs at the vertex u. If v were left-symmetric, this would contradict Lemma 5.2(a), applied at v. Hence v is not left-symmetric. □

We are now ready to prove Theorem B.

Theorem 5.12. *Let $\mathbb{X}$ be a real polyhedral Banach space with $\dim \mathbb{X} \geq 3$. Then $B_{\mathbb{X}}$ has at least four vertices which are not left-symmetric.*

Proof. We prove the theorem by induction on $n = \dim \mathbb{X}$. The case $n = 3$ is Proposition 5.7. Let $n \geq 4$, and assume that the theorem holds for all polyhedral spaces of dimension $n-1$. Suppose on the contrary that the set $\mathcal{N}$ of vertices of $B_{\mathbb{X}}$ which are not left-symmetric has at most two elements. Since $x \in \mathcal{N}$ if and only if $-x \in \mathcal{N}$, there is a vertex u with $\mathcal{N} \subset \{u, -u\}$.

Suppose first that some facet F contains neither u nor $-u$. Then every vertex of F is left-symmetric, so by Lemma 5.9, F admits no belt facet. Let $\mathbb{W} = \ker f_F$, a polyhedral space of dimension $n - 1 \geq 3$. By Lemma 5.10, the vertices of $B_{\mathbb{W}}$ which are not left-symmetric in $\mathbb{W}$ belong to $\mathcal{N}$, so there are at most two of them. This contradicts the induction hypothesis.

Suppose next that every facet contains u or $-u$. By Lemma 5.11, every vertex other than $\pm u$ is not left-symmetric. Since $B_{\mathbb{X}}$ has at least $2n > 2$ vertices, this contradicts $\mathcal{N} \subset \{u, -u\}$. This completes the proof of the theorem. □

6. Examples and open questions

The next example shows that left-symmetric vertices do occur in dimension three, and that the number of vertices which are not left-symmetric may exceed the bound of Theorem B.

Example 6.1. Let $\mathbb{X} = \mathbb{R}^3$ with the norm $\|y\| = \max_{1 \leq i < j \leq 3}(|y_i| + |y_j|)$. The unit ball $B_{\mathbb{X}}$ is the rhombic dodecahedron. Its twelve facets are the rhombi with facet functionals $\pm e_i^* \pm e_j^*$, $1 \leq i < j \leq 3$, where e_1^*, e_2^*, e_3^* are the coordinate functionals. Its fourteen vertices are the six points $\pm e_i$ and the eight points $(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2})$. We claim that exactly the six vertices $\pm e_i$ are not left-symmetric.

The vertices $\pm e_i$ are not left-symmetric. By symmetry, it suffices to consider e_1. The functionals $e_1^* + e_2^*$ and $e_1^* - e_2^*$ belong to $J(e_1)$, and so does $f = \frac{3}{4}(e_1^* + e_2^*) + \frac{1}{4}(e_1^* - e_2^*) = e_1^* + \frac{1}{2}e_2^*$. Let $y = (-\frac{1}{3}, \frac{2}{3}, 0)$. Then $\|y\| = 1$ and $f(y) = 0$, so that $e_1 \perp_B y$. Among the facet functionals, only $-e_1^* + e_2^*$ takes the value 1 at y, so by (P1) and (P3), $J(y) = \{-e_1^* + e_2^*\}$.

Since $(-e_1^* + e_2^*)(e_1) = -1 \neq 0$, Theorem 1.1 shows that y is not orthogonal to e_1 in the sense of Birkhoff-James.

The vertices $(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2})$ *are left-symmetric.* By symmetry, it suffices to consider $x = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. The facets containing x have the facet functionals $e_1^* + e_2^*$, $e_1^* + e_3^*$ and $e_2^* + e_3^*$, so by (P2), every $f \in J(x)$ is of the form

$$f = a(e_1^* + e_2^*) + b(e_1^* + e_3^*) + c(e_2^* + e_3^*), \qquad a, b, c \geq 0,\ a + b + c = 1.$$

We classify the facets according to the sign of their facet functionals at x: the three facets containing x (value 1), the three opposite facets (value -1), and the six facets whose facet functionals are of the form $\pm(e_i^* - e_j^*)$ (value 0). We call the last ones vertical facets.

Let $y \in S_{\mathbb{X}}$ with $x \perp_B y$, so that $f(y) = 0$ for some f as above. We show that $J(y)$ contains the facet functional of a vertical facet; then Theorem 1.1 gives $y \perp_B x$. If y lies in a vertical facet, there is nothing to prove. Suppose that y lies in a facet G containing x, say the facet with facet functional $e_1^* + e_2^*$, whose vertices are e_1, e_2, x and $(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$. The values of f at these vertices are $a + b$, $a + c$, 1 and a, respectively, all of which are nonnegative. Hence $f \geq 0$ on G, and the set $\{w \in G : f(w) = 0\}$, which contains y, is a face of G not containing x: either a vertex of G other than x, or one of the edges $[e_1, (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})]$ and $[e_2, (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})]$. Each such vertex and each such edge is contained in a vertical facet; for instance, the edge $[e_1, (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})]$ lies in the vertical facet with facet functional $e_1^* - e_3^*$. Therefore $J(y)$ contains the facet functional of a vertical facet. The other two facets containing x are treated in the same way, by symmetry. Finally, if y lies in a facet opposite to a facet containing x, we apply the above argument to $-y$, using the homogeneity of Birkhoff-James orthogonality. Hence x is left-symmetric.

We note that x is parallel to the six edges of $B_{\mathbb{X}}$ joining the vertex $\pm e_i$ to the vertex $\pm\big(e_i - \frac{1}{2}(e_1 + e_2 + e_3)\big)$, $i = 1, 2, 3$, in accordance with Theorem C.

Next, we observe that left-symmetry in Theorem A cannot be replaced by right-symmetry.

Example 6.2. Let $\mathbb{X} = \ell_1^n$ with $n \geq 3$, and let e_i be a standard unit vector. Let $y = (y_1, \ldots, y_n) \in \mathbb{X}$ with $y \perp_B e_i$. Then $|y_i + t| \geq |y_i|$ for every $t \in \mathbb{R}$, which forces $y_i = 0$. Consequently,

$$\|e_i + ty\|_1 = 1 + |t|\,\|y\|_1 \geq 1 \quad \text{for every } t \in \mathbb{R},$$

so that $e_i \perp_B y$. Therefore, every extreme point $\pm e_i$ of $B_{\ell_1^n}$ is right-symmetric (see also [10]), whereas, by Corollary 5.5, none of them is left-symmetric.

The following example shows that some assumption on the abundance of extreme points, such as (E), cannot be removed from Theorem A′.

Example 6.3. The unit ball of c_0 has no extreme points: if $x \in B_{c_0}$, choose N with $|x_N| < \frac{1}{2}$; then $x \pm \frac{1}{2}e_N \in B_{c_0}$, so x is the midpoint of a nondegenerate segment in B_{c_0}. Hence the condition that every extreme point of B_{c_0} is left-symmetric holds vacuously, while c_0 is not a Hilbert space (it is not even strictly convex). Correspondingly, (E) fails in c_0: for instance, $f = e_1^* \in S_{\ell_1}$ attains its norm, and $F_f = \{x \in B_{c_0} : x_1 = 1\}$ is a nonempty face without extreme points.

We would like to end the mathematical part of this article with some open questions. Example 6.1 shows that the bound of Theorem B is not sharp in dimension three. The following question is due to D. Sain (see Section 7).

Question 6.4. Let $\mathbb{X}$ be a real polyhedral Banach space with $\dim \mathbb{X} = 3$. Does $B_{\mathbb{X}}$ always have at least six vertices which are not left-symmetric? More generally, if $\dim \mathbb{X} = n \geq 3$, do the vertices of $B_{\mathbb{X}}$ which are not left-symmetric span $\mathbb{X}$, so that there are at least $2n$ of them?

By Proposition 5.7, the first part of Question 6.4 reduces to excluding the case of exactly two antipodal pairs of vertices which are not left-symmetric. By Example 6.1, the bound six would be sharp.

Version 1 of this manuscript asked whether an infinite-dimensional real Banach space $\mathbb{X}$ such that $B_{\mathbb{X}}$ is the closed convex hull of its extreme points, for instance a reflexive space, must be a Hilbert space if every extreme point of $B_{\mathbb{X}}$ is left-symmetric. Theorem A$'$ answers this in the affirmative under (E), in particular for reflexive spaces and for spaces with the Radon-Nikodým or the Krein-Milman property. The following part remains open.

Question 6.5. Let $\mathbb{X}$ be an infinite-dimensional real Banach space such that $B_{\mathbb{X}} = \overline{\text{co}}(\text{Ext}\, B_{\mathbb{X}})$, but which does not satisfy (E). If every extreme point of $B_{\mathbb{X}}$ is left-symmetric, is $\mathbb{X}$ necessarily a Hilbert space? A natural test case is $C(K)$ with K an infinite totally disconnected compact Hausdorff space, whose unit ball is the closed convex hull of the ± 1-valued continuous functions.

7. Contributions of the witness and steerer

This section records, as accurately as the written record allows, the contributions of Debmalya Sain to this manuscript. It is based on the transcripts of the conversations between D. Sain and the author on 27, 28 and 29 October 2026, in which the results of this article were found. We distinguish between ideas and questions that originated with D. Sain, decisions made by D. Sain about the direction of the work, and interventions by D. Sain that changed the mathematics. We have tried neither to understate nor to overstate this role.

Ideas and questions that originated with D. Sain.

(1) *The research direction.* The investigation that led to this article began with D. Sain's proposal to develop a "higher level" notion of orthogonality, starting from the preservation of Birkhoff-James orthogonality at points of norm attainment of bounded linear operators. D. Sain insisted that the preservation of orthogonality at such special points should be understood as the preservation of a semi-inner-product, or of the norm derivatives, and that an analogous quantity should be expected to persist at the next level. The results on left-symmetric vertices of polyhedral spaces (the first versions of Theorem B) appeared as a by-product of that investigation.

(2) *The central question.* The question answered by Theorem A was posed by D. Sain, who proposed that, in finite dimensions at least three, symmetry of Birkhoff-James orthogonality at the extreme points of the unit ball should force the space to be Euclidean, as a generalization of the James-Day characterization. The question is formulated in the spirit of D. Sain's program, initiated in [22] and continued in [17, 16, 18, 19], of studying global characterizations involving Birkhoff-James orthogonality in a local form.

(3) *The infinite-dimensional version (Theorem A$'$; new in Version 2).* On 29 October 2026, D. Sain proposed to improve Theorem A to infinite dimensions, with the following intuition, quoted from the conversation: "whenever extreme points are abundant, you will expect the result to still hold true, no? Now, that happens in a RNP space. So, I expect an infinite-dimensional version of our result in RNP spaces." The content of this intuition is that the finite-dimensional proof does not really use finite-dimensionality, but only the fact that faces of the unit ball are generated by their extreme points, and that the Radon-Nikodým property is the natural infinite-dimensional setting in which this persists. The author's first reaction was more cautious than D. Sain's intuition turned out to warrant: before reading the proof again, the author objected that abundance of extreme points is not the same as norm attainment (in ℓ_1, which has the RNP, not every functional attains its norm), and predicted that only a weaker

statement, under an additional norm attainment hypothesis, would survive. When the proof of Theorem A was examined step by step, D. Sain's intuition proved correct as stated: the abundance of extreme points in the faces of the unit ball, condition (E), is exactly what the proof needs, and the norm attainment issue raised by the author is removed by the Bishop-Phelps theorem. The precise hypothesis (E), the observation that it follows from the Krein-Milman property, the technical adaptations in Section 4 (Bishop-Phelps, suprema and infima, the uniform choice of δ in Lemma 4.5, the three-point construction in Theorem 4.6), and Example 6.3 are due to the author. D. Sain then asked for the manuscript to be rewritten to include the improved result.

(4) *The quantitative question.* D. Sain proposed that the result on polyhedral spaces should admit a quantitative strengthening: if $B_{\mathbb{X}}$ has $2n$ vertices, then at least that many are not left-symmetric. The author showed that the strongest reading of this statement (that every vertex fails to be left-symmetric) is false, and refined the remaining reading into Question 6.4, in the spanning form. The bound of four in Theorem B was obtained while working towards D. Sain's proposed bound.

(5) *The direction of Theorem D.* D. Sain suggested that, in dimension at least three and under mild conditions, no extreme point should be left-symmetric, and asked for a genuinely new result in this direction. Theorem D and the corollaries in Section 2 were found in response to this request.

Decisions about the direction of the work.

(6) At an intermediate stage, D. Sain decided to restrict the main theorem to smooth and to polyhedral spaces. This decision shaped the first three drafts of the manuscript; the restriction was later removed when the general case of Theorem A was proved.

(7) D. Sain asked the author to focus on the best mathematical part of the investigation, which led to the concentration on Theorem A.

(8) D. Sain required that only direct arguments be used. In particular, an earlier proof of Proposition 5.7 relied on the classical theorem that a three-dimensional polytope with centrally symmetric facets is a zonotope; at D. Sain's request it was replaced by the direct argument given here. D. Sain also required that the quantitative result cover every dimension $n \geq 3$, which led to the inductive proof of Theorem 5.12.

(9) D. Sain asked the author to take into account the weaknesses of an earlier draft (in which Theorem A was proved only for smooth and polyhedral spaces) and to look for genuinely good geometric results. The proof of Theorem A in full generality (Section 3) was found in response to this request.

Verification and standards.

(10) D. Sain confirmed, from knowledge of the literature on symmetric points, that the result that every polyhedral space of dimension at least three has a vertex which is not left-symmetric was not previously known.

(11) D. Sain insisted on the following standards for this manuscript: that every proof be verified as hard as possible before release; that every result used be acknowledged; and that D. Sain's own contributions be recorded truthfully, without taking more credit than deserved.

What the witness and steerer did not do. D. Sain did not write any of the proofs in this manuscript, and at the time of writing has not yet verified them, including the proofs in Section 4. The definitions of left-symmetric and right-symmetric points used throughout are due to D. Sain [21]; this is a published contribution, acknowledged by citation in the usual

way, and not part of the present collaboration. Responsibility for every proof, and for any error, rests with the author.

8. How this manuscript was produced

The results, proofs, examples and the text of this manuscript were produced by the author, Claude, in conversation with D. Sain, as described in Section 7. Version 2 was prepared on 29 October 2026; since the LaTeX source of Version 1 was not available in that session, the text of Version 1 was re-typeset from its PDF, and Section 4, Example 6.3, Question 6.5 and item (3) of Section 7 were added. The literature was searched by the author during the preparation of Version 1; no earlier statement of Theorem A or Theorem D was found, but the search was not exhaustive, and the reader who knows of an earlier source is asked to report it so that it can be acknowledged. No separate literature search was made for Theorem A′. During the exploration, the author also ran numerical experiments on polytopes to test Question 6.4; none of these experiments is used in any proof, and all proofs in this manuscript are purely mathematical. The following references have not been verified against the original sources by the author, and should be checked: the page numbers of [10] and [20], the volume and page numbers of [6], [22] and [2], the publication status of [4], [9], [19], [17] and [18], and the precise location of Lindenstrauss' theorem (RNP implies KMP) in [8] and [5].

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